

WHICH OF THE FIVE BOROUGHS COMPRISING NEW YORK CITY ARE MOST SIMILAR?

Coursera Applied Data Science, Capstone
Course

Daniel Costelloe

April 11, 2019



INTRODUCTION:

NEW YORK CITY IS COMPRISED OF FIVE BOROUGHES, ANALOGOUS TO COUNTIES, WHICH CAME TOGETHER FORMING NEW YORK CITY.

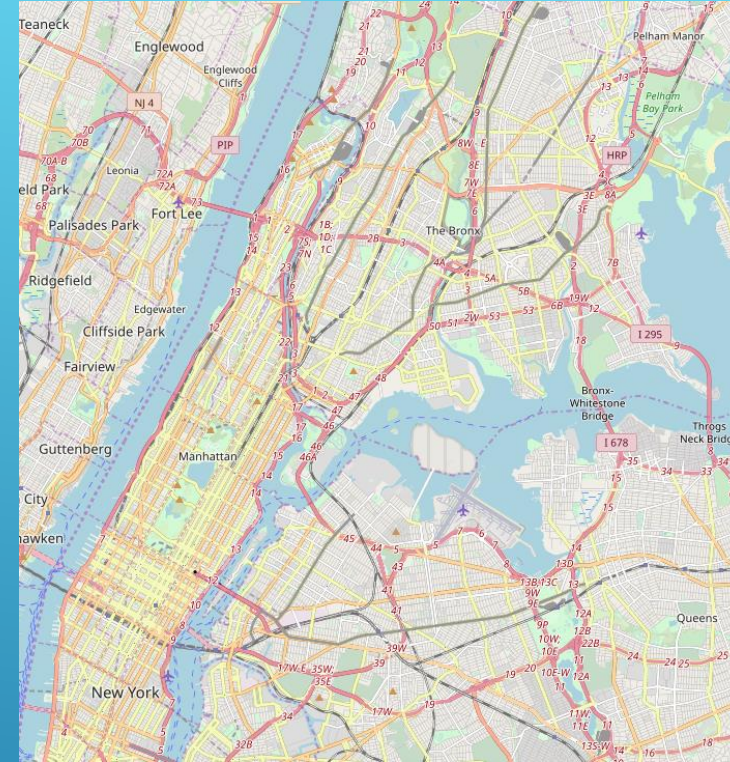
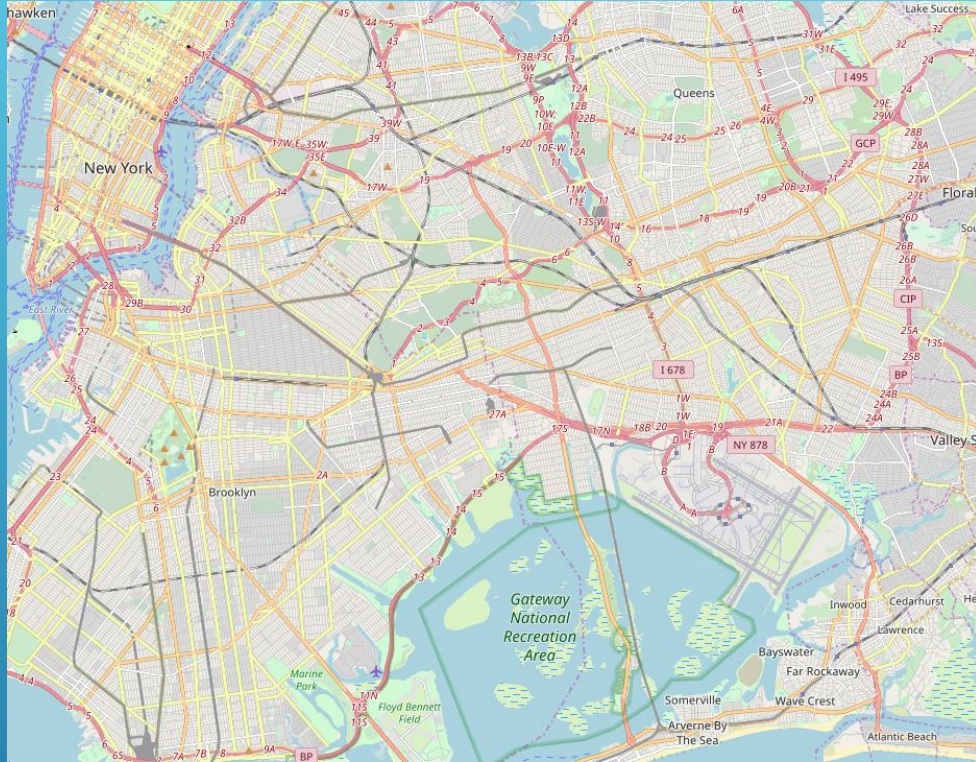
The five boroughs were not previously each a city, some were parts of other counties. But for the purpose of this presentation, can be thought of as five cities comprising New York City.

► The five boroughs are:

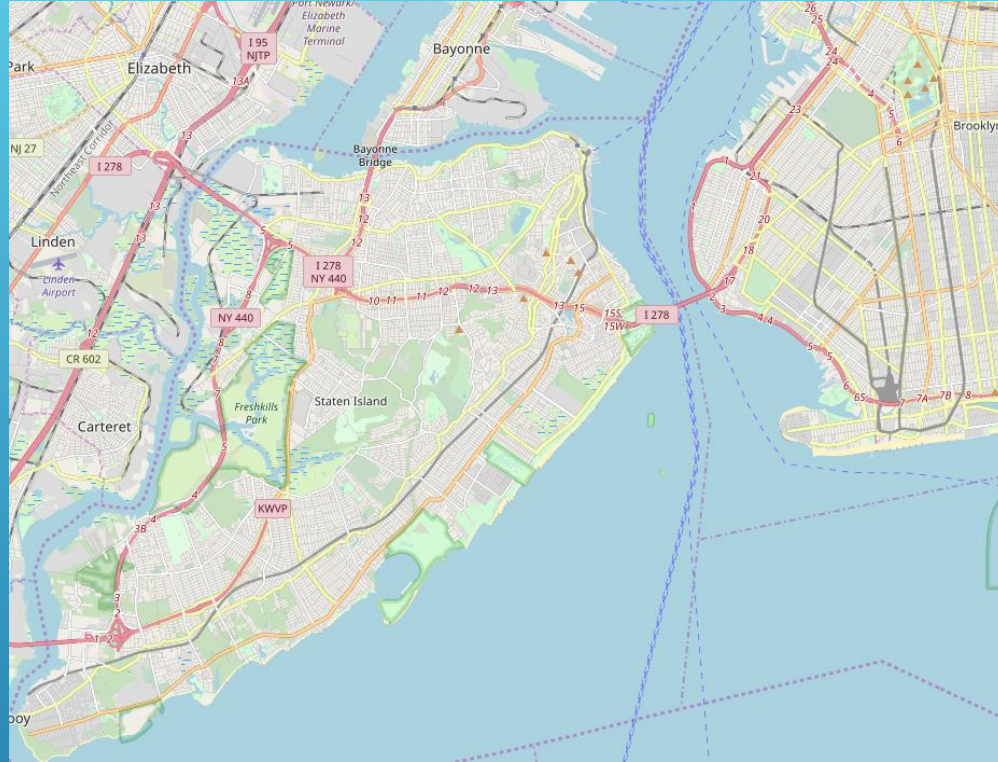
1. Bronx
2. Brooklyn
3. Manhattan
4. Queens
5. Staten Island

THE QUESTION IS: WHICH BOROUGHES
ARE MOST SIMILAR?

Several white lines of varying lengths and slopes are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.



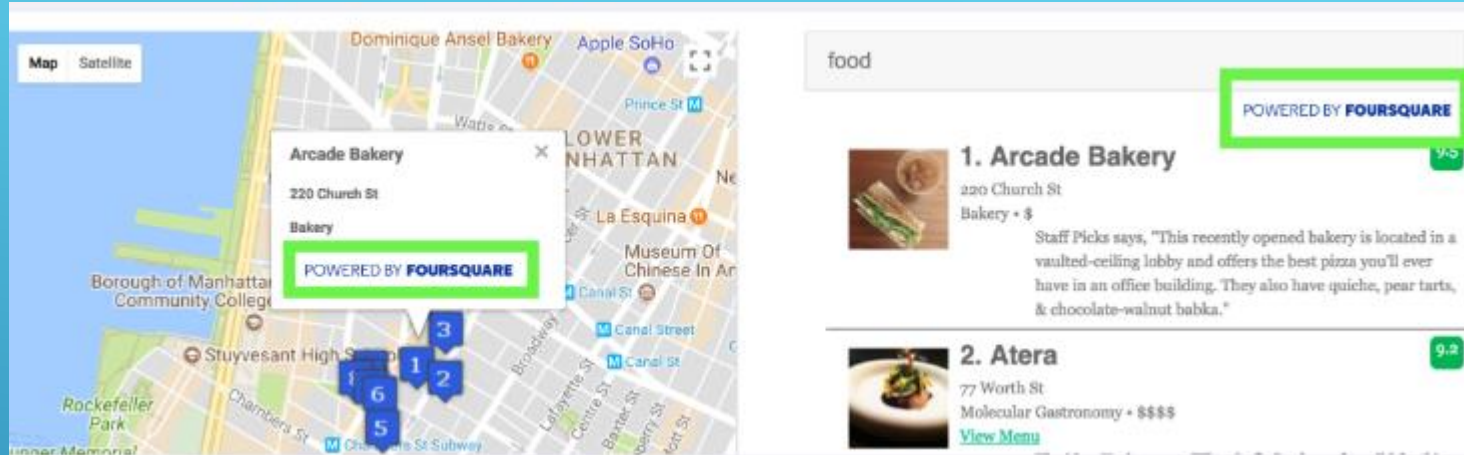
GEOGRAPHICALLY BROOKLYN AND QUEENS ARE CONTIGUOUS. THE BRONX AND MANHATTAN ARE VERY CLOSE ON EITHER SIDE OF THE HARLEM RIVER



GEOGRAPHICALLY STATEN ISLAND IS CLOSER TO NEW JERSEY THAN IT IS TO ANY OF THE BOROUGHES. IT IS CONNECTED TO BROOKLYN BY A BRIDGE.

SOURCES OF DATA

1. FourSquare



FourSquare provides geographically based venue data accessed by an API call.

2. NYU Faculty Digital Archive, FDACommunities & Collections, Division of Libraries, Spatial Data Repository, 2014 New York City Neighborhood Names

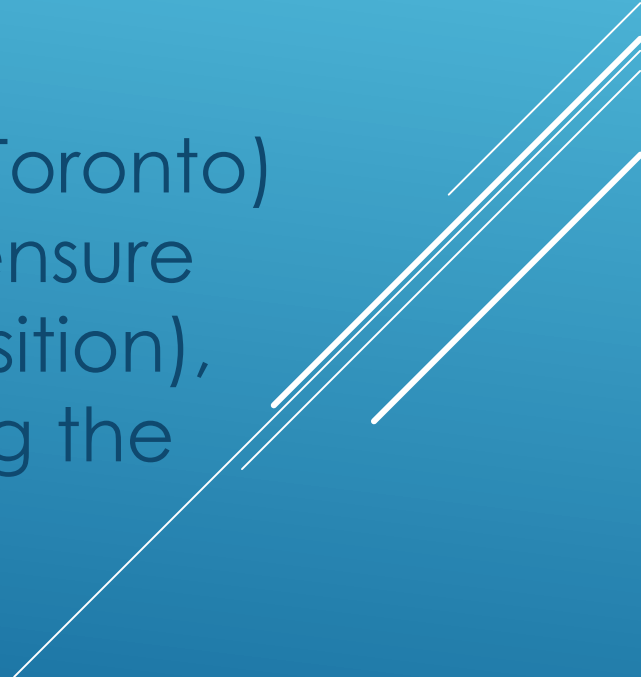
(<http://hdl.handle.net/2451/34572>) Names and coordinates of NYC neighborhoods.

3. Python geolocator.geocode

Documented at <https://geopy.readthedocs.io/en/stable/>

Used to get coordinates of locations, for example New York City.

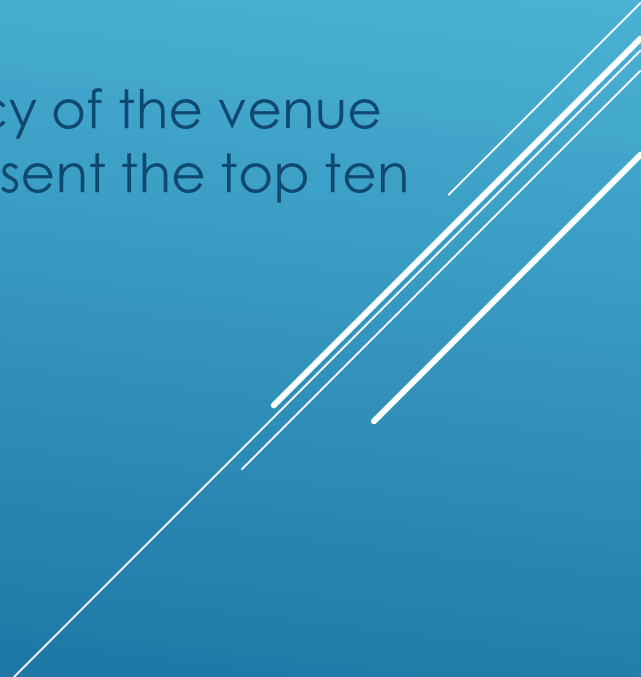
DATA CLEANING

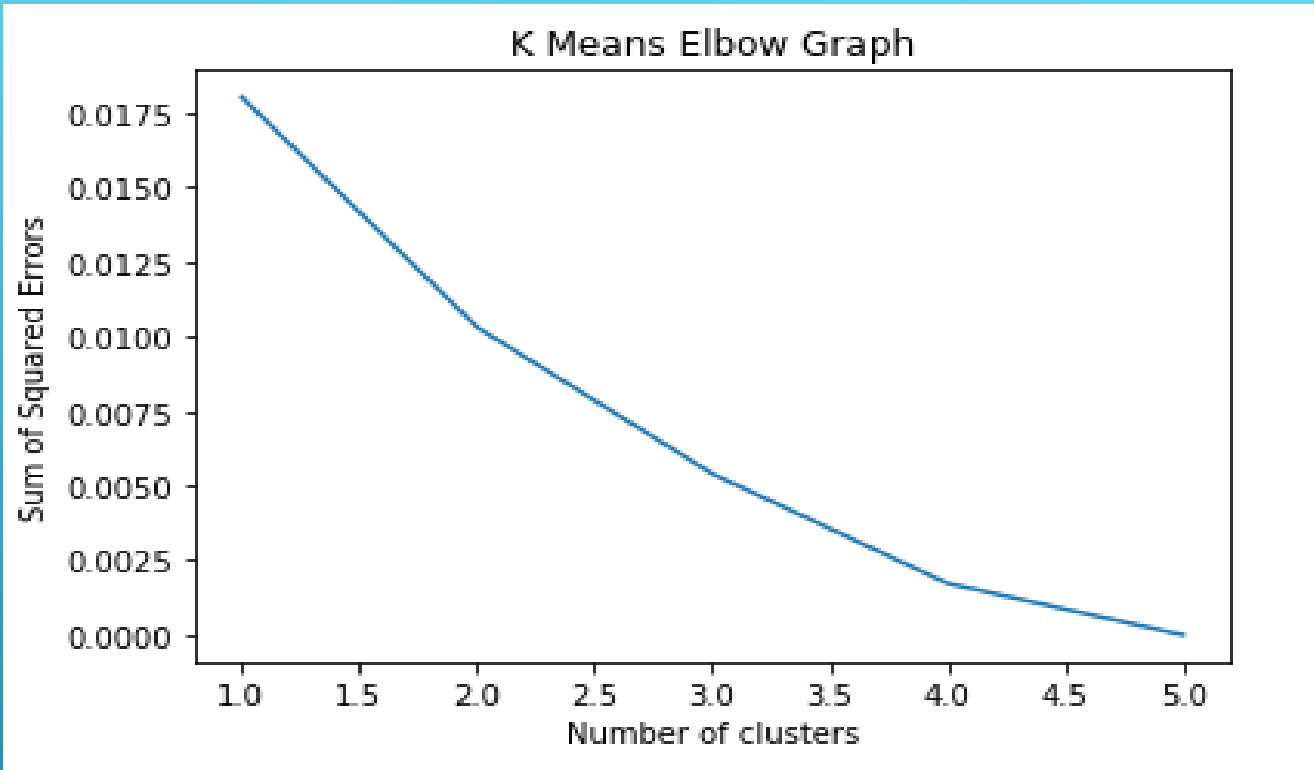
1. Neighborhood names were made unique by appending the borough name to the neighborhood name.
 2. In another use of FourSquare venue data (for Toronto) a venue category included Neighborhood. To ensure unique column names in the data (after transposition), the venue category was changed by appending the word Category to Neighborhood.
- 
- Several white lines of varying lengths and angles are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.

METHODOLOGY:

K-Means clustering was used to group the boroughs based on the relative frequency of venue types within boroughs.

Numerical variables were created to represent the relative frequency of the venue categories within each borough. These variables were sorted to present the top ten venue categories for each borough.

Several white lines of varying lengths and slopes are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.



ALTHOUGH AN “ELBOW” GRAPH WAS LOOKED AT, THE CHOICE OF THE NUMBER OF CLUSTERS WAS DRIVEN BY WHAT WAS BEST FOR THIS PARTICULAR QUESTION OF GROUPING THE BOROUGHs. PUTTING ALMOST EACH BOROUGH IN ITS OWN CLUSTER, OR PUTTING MOST BOROUGHs IN ONE CLUSTER WOULD NOT TELL US WHICH BOROUGHs ARE MOST SIMILAR. A GOLDBLOCKS, JUST RIGHT, NUMBER OF THREE CLUSTERS WAS CHOSEN FOR THIS REASON.

Borough	Number_1_Venue_Type	Number_2_Venue_Type	Number_3_Venue_Type	Number_4_Venue_Type	Number_5_Venue_Type	Number_6_Venue_Type	Number_7_Venue_Type	Number_8_Venue_Type	Number_9_Venue_Type	Number_10_Venue_Type
Bronx	Pizza Place	Deli / Bodega	Supermarket	Pharmacy	Chinese Restaurant	Donut Shop	Bus Station	Sandwich Place	Italian Restaurant	Spanish Restaurant
Brooklyn	Cluster Labels	Pizza Place	Coffee Shop	Bar	Italian Restaurant	Grocery Store	Café	Deli / Bodega	Bakery	Ice Cream Shop
Manhattan	Cluster Labels	Coffee Shop	Park	American Restaurant	Italian Restaurant	Café	Pizza Place	Bakery	Gym	Cocktail Bar
Queens	Pizza Place	Deli / Bodega	Chinese Restaurant	Bakery	Pharmacy	Donut Shop	Bank	Italian Restaurant	Sandwich Place	Coffee Shop
Staten Island	Cluster Labels	Pizza Place	Bus Stop	Deli / Bodega	Italian Restaurant	Bagel Shop	Pharmacy	Bank	Sandwich Place	Chinese Restaurant

EACH BOROUGH SHOWING ITS MOST
FREQUENT VENUE TYPES

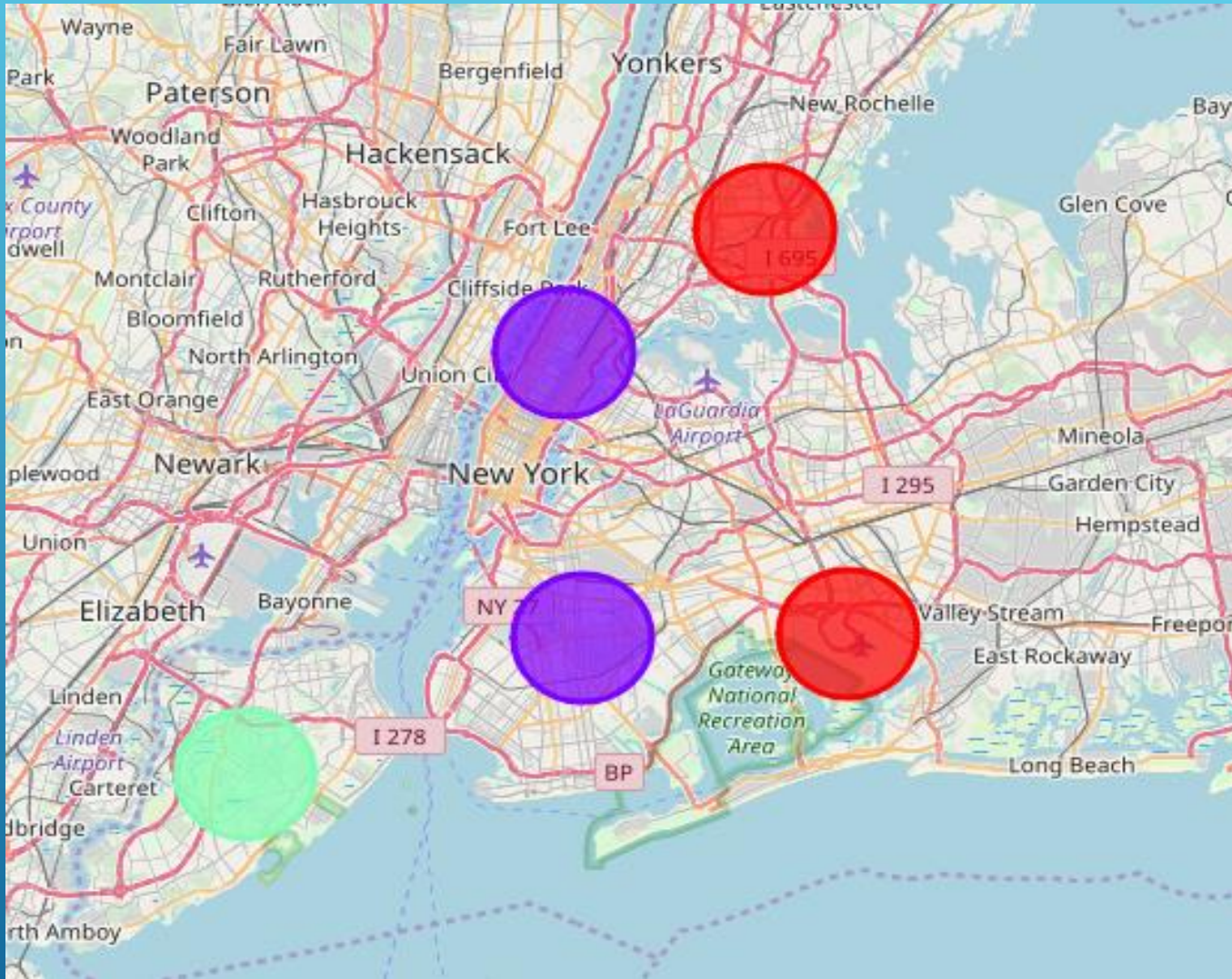
Cluster_Name	Borough	Number_1_Venue_Type	Number_2_Venue_Type	Number_3_Venue_Type	Number_4_Venue_Type	Number_5_Venue_Type	Number_6_Venue_Type	Number_7_Venue_Type	Number_8_Venue_Type	Number_9_Venue_Type
Bronx and Queens (Pizza Place and Deli / Bodega)	Bronx	Pizza Place	Deli / Bodega	Supermarket	Pharmacy	Chinese Restaurant	Donut Shop	Bus Station	Sandwich Place	Italian Restaurant
Bronx and Queens (Pizza Place and Deli / Bodega)	Queens	Pizza Place	Deli / Bodega	Chinese Restaurant	Bakery	Pharmacy	Donut Shop	Bank	Italian Restaurant	Sandwich Place
Brooklyn and Manhattan (Coffee Shop)	Brooklyn	Pizza Place	Coffee Shop	Bar	Italian Restaurant	Grocery Store	Café	Deli / Bodega	Bakery	Ice Cream Shop
Brooklyn and Manhattan (Coffee Shop)	Manhattan	Coffee Shop	Park	American Restaurant	Italian Restaurant	Café	Bakery	Pizza Place	Cocktail Bar	Gym
Staten Island (Pizza Place and Bodega)	Staten Island	Pizza Place	Bus Stop	Deli / Bodega	Italian Restaurant	Bagel Shop	Pharmacy	Sandwich Place	Bank	Grocery Store

THE BOROUGHs WERE CLUSTERED AS:

1. THE BRONX AND QUEENS
2. BROOKLYN AND MANHATTAN
3. STATEN ISLAND

CLUSTER NAMES WERE THEN ASSIGNED BASED ON THE INCLUDED BOROUGHs AND THE MOST COMMON TYPE OF VENUES

CLOROPLETH MAP SHOWING THE CLUSTERS.



DISCUSSION:

The boroughs clustered together were non-continuous.

The Bronx and Queens were the most similar boroughs. Not only did they match on their first two categories, but within the next four categories (Pizza and Deli), they both had: Chinese Restaurant, Pharmacy, and Donut Shop.

Staten Island stood out for having Bus Stop in its top two categories and Bagel Shop in its top five.

Brooklyn and Manhattan both had Coffee Shop in their top two, Italian Restaurant as fourth, and Café ranked as either fifth or sixth.

Several white lines of varying lengths and angles are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.

SUGGESTIONS FOR FURTHER RESEARCH

Further research could look at grouping ethnic restaurants together by continent or general geographic location. This might allow more restaurant categories to enter the most frequent venue types.

Coffee Shop and Café could be combined.

Bus Stop might be eliminated as a category, since subway stops are not considered categories.

Several white lines of varying lengths and angles are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.

CONCLUSIONS

People who would like to live in Manhattan, but are deterred by the high cost of living in Manhattan might consider Brooklyn.

People who like and know either the Bronx or Queens should become more familiar with the other borough, and consider it if looking to relocate.

People who do not like the other boroughs should explore Staten Island for its distinctive quality.

Several white lines of varying lengths and thicknesses are positioned diagonally in the bottom right corner of the slide, creating a modern, abstract graphic element.